WEEK 9 REPORT

HOUSING DATA CLEANING & TRANSFORMATION

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1. Introduction

Housing Data Cleaning and Transformation is an important process in data analysis. In this task, the King County House Sales Dataset is used to analyze house prices and property characteristics. Power BI Power Query is used to clean the data, transform date columns, and create new columns for better analysis. This process helps prepare the dataset for accurate and meaningful insights.

2. Problem Statement

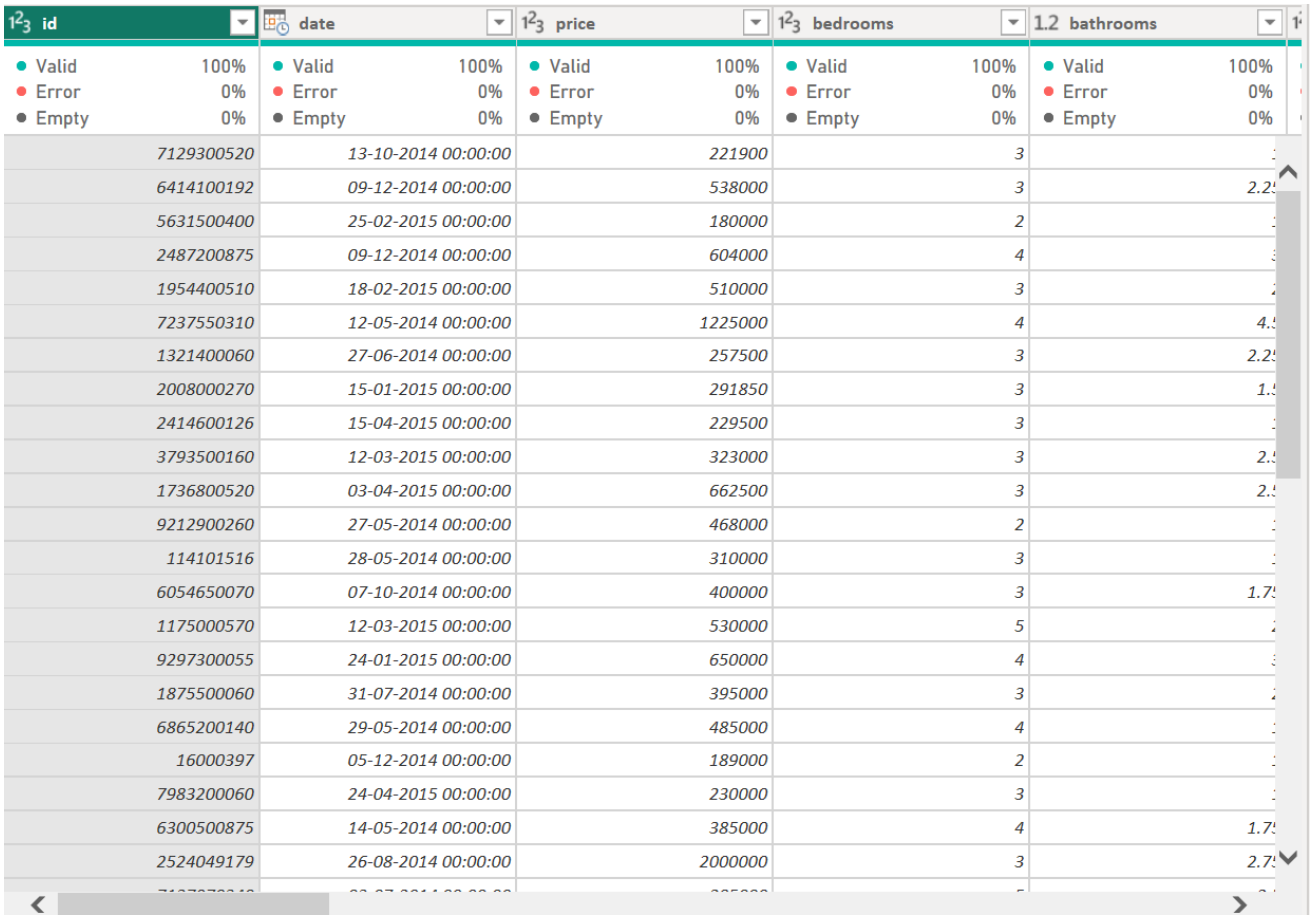
invalid A real-estate organization wants to analyze historical house sales data to understand property characteristics and the factors that influence house prices. The dataset must be cleaned and transformed to handle records and derive useful information such as property age, price per square foot, total area, renovation status, property size, and price category.

3. Screenshots and Explanations

The following section contains screenshots of the data cleaning and transformation steps performed in Power Query. Insert each screenshot below its matching heading.

Screenshot 1: Housing Dataset in Power Query Editor

The King County House Sales dataset was imported into Power Query Editor. The dataset contains information about house sales and property characteristics, such as sale date, price, bedrooms, bathrooms, and living area.



id	date	price	bedrooms	bathrooms
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
7129300520	13-10-2014 00:00:00	221900	3	1
6414100192	09-12-2014 00:00:00	538000	3	2.25
5631500400	25-02-2015 00:00:00	180000	2	1
2487200875	09-12-2014 00:00:00	604000	4	3
1954400510	18-02-2015 00:00:00	510000	3	2
7237550310	12-05-2014 00:00:00	1225000	4	4.5
1321400060	27-06-2014 00:00:00	257500	3	2.25
2008000270	15-01-2015 00:00:00	291850	3	1.5
2414600126	15-04-2015 00:00:00	229500	3	1
3793500160	12-03-2015 00:00:00	323000	3	2.5
1736800520	03-04-2015 00:00:00	662500	3	2.5
9212900260	27-05-2014 00:00:00	468000	2	1
114101516	28-05-2014 00:00:00	310000	3	1
6054650070	07-10-2014 00:00:00	400000	3	1.75
1175000570	12-03-2015 00:00:00	530000	5	2
9297300055	24-01-2015 00:00:00	650000	4	3
1875500060	31-07-2014 00:00:00	395000	3	2
6865200140	29-05-2014 00:00:00	485000	4	1
16000397	05-12-2014 00:00:00	189000	2	1
7983200060	24-04-2015 00:00:00	230000	3	1
6300500875	14-05-2014 00:00:00	385000	4	1.75
2524049179	26-08-2014 00:00:00	2000000	3	2.75

Screenshot 2: Extracting Sale Year and Sale Month

The sale date was used to create Sale Year and Sale Month columns. These columns help analyze house sales by year

	long		sqft_living15		sqft_lot15		Sale Year		Sale Month	
	100% Valid	100%	100% Valid	100%	100% Valid	100%	100% Valid	100%	100% Valid	100%
	0% Error	0%	0% Error	0%	0% Error	0%	0% Error	0%	0% Error	0%
	0% Empty	0%	0% Empty	0%	0% Empty	0%	0% Empty	0%	0% Empty	0%
1	47.5112	-122.257	1340		5650		2014		10	
2	47.721	-122.319	1690		7639		2014		12	
3	47.7379	-122.233	2720		8062		2015		2	
4	47.5208	-122.393	1360		5000		2014		12	
5	47.6168	-122.045	1800		7503		2015		2	
6	47.6561	-122.005	4760		101930		2014		5	
7	47.3097	-122.327	2238		6819		2014		6	
8	47.4095	-122.315	1650		9711		2015		1	
9	47.5123	-122.337	1780		8113		2015		4	
10	47.3684	-122.031	2390		7570		2015		3	
11	47.6007	-122.145	2210		8925		2015		4	
12	47.69	-122.292	1330		6000		2014		5	
13	47.7558	-122.229	1780		12697		2014		5	
14	47.6127	-122.045	1370		10208		2014		10	
15	47.67	-122.394	1360		4850		2015		3	
16	47.5714	-122.375	2140		4000		2015		1	
17	47.7277	-121.962	1890		14018		2014		7	
18	47.6648	-122.343	1610		4300		2014		5	
19	47.3089	-122.21	1060		5095		2014		12	
20	47.3343	-122.306	1280		8850		2015		4	
21	47.7025	-122.341	1400		4980		2014		5	
22	47.5316	-122.233	4110		20336		2014		8	

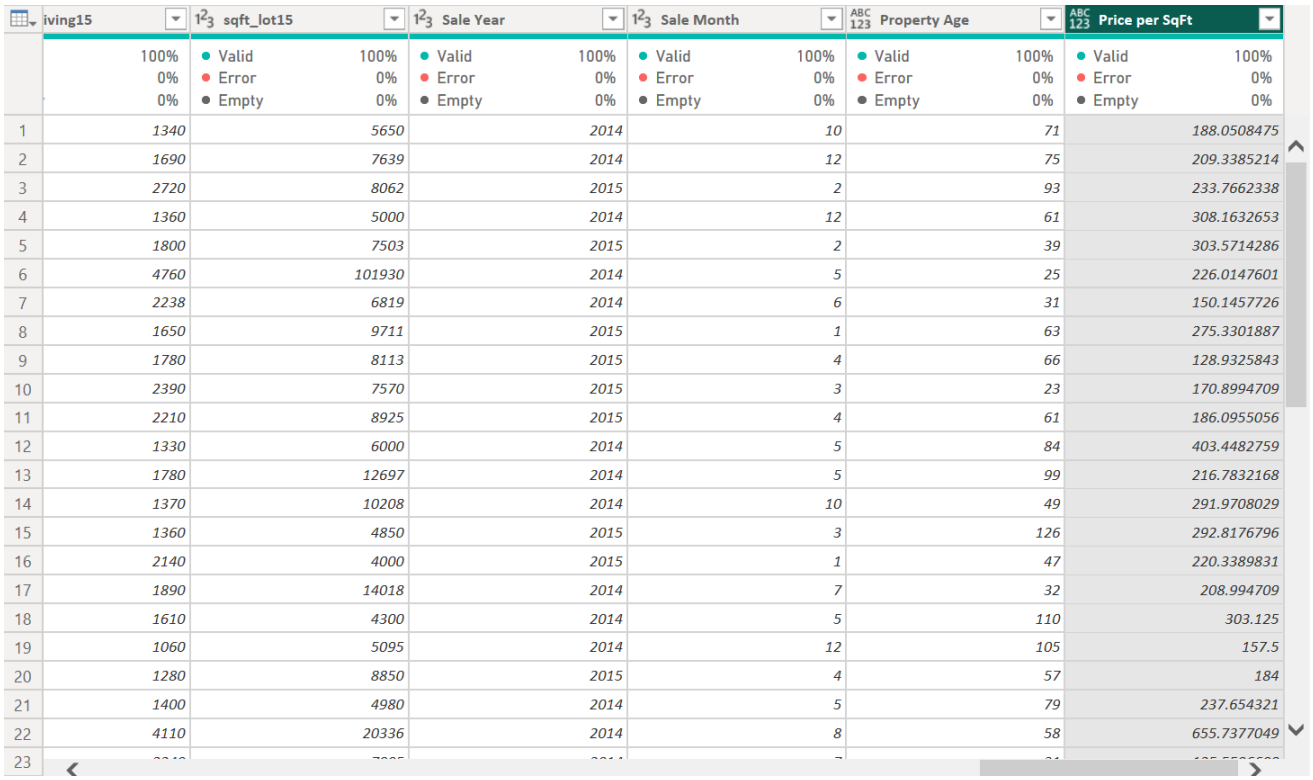
Screenshot 3: Calculating Property Age

A custom column named Property Age was created to show the age of a property at the time of sale. This helps compare properties based on their age.

	sqft_living15		sqft_lot15		Sale Year		Sale Month		Property Age	
100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
0%	Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%
-122.257		1340		5650		2014		10		71
-122.319		1690		7639		2014		12		75
-122.233		2720		8062		2015		2		93
-122.393		1360		5000		2014		12		61
-122.045		1800		7503		2015		2		39
-122.005		4760		101930		2014		5		25
-122.327		2238		6819		2014		6		31
-122.315		1650		9711		2015		1		63
-122.337		1780		8113		2015		4		66
-122.031		2390		7570		2015		3		23
-122.145		2210		8925		2015		4		61
-122.292		1330		6000		2014		5		84
-122.229		1780		12697		2014		5		99
-122.045		1370		10208		2014		10		49
-122.394		1360		4850		2015		3		126
-122.375		2140		4000		2015		1		47
-121.962		1890		14018		2014		7		32
-122.343		1610		4300		2014		5		110
-122.21		1060		5095		2014		12		105
-122.306		1280		8850		2015		4		57
-122.341		1400		4980		2014		5		79
-122.233		4110		20336		2014		8		58

Screenshot 4: Calculating Price per Square Foot

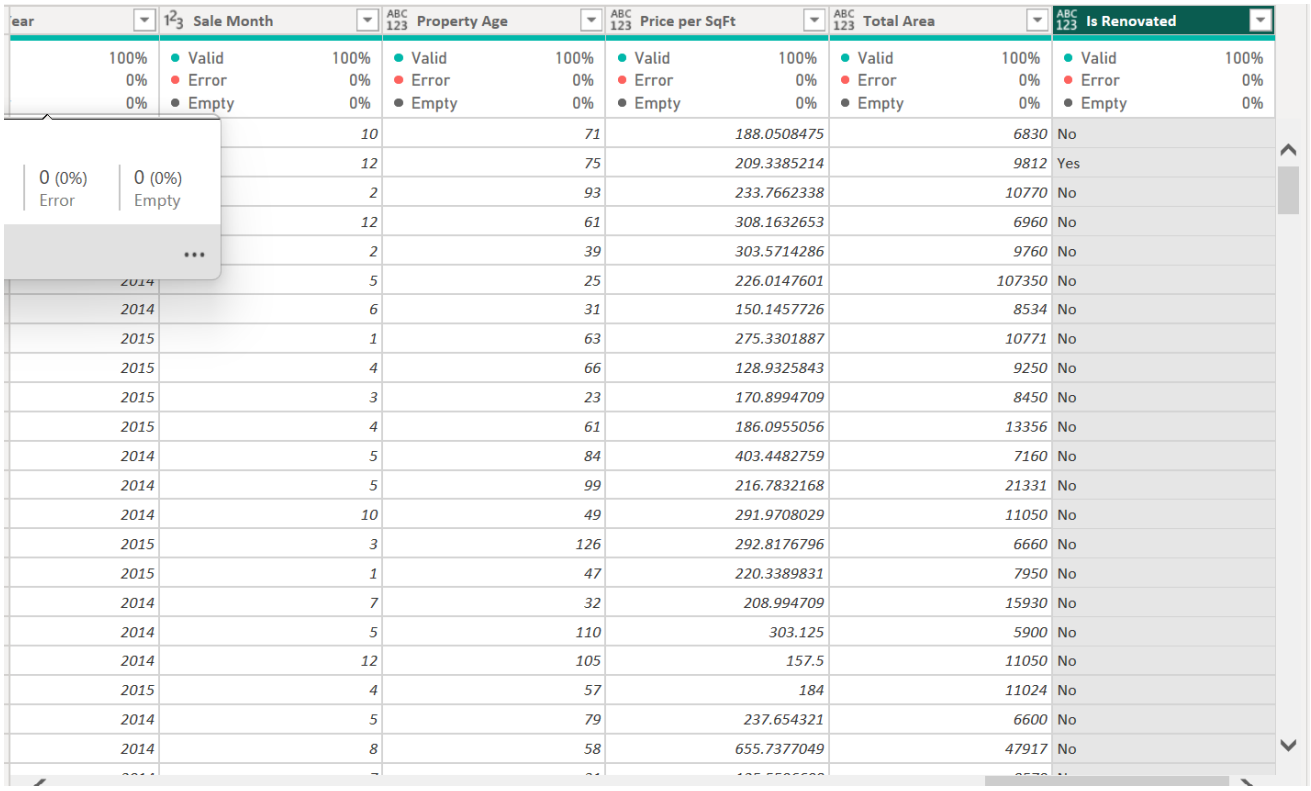
The Price per SqFt column was created by dividing the sale price by the living area. This provides a way to compare property prices relative to their size.



	iving15	sqft_lot15	Sale Year	Sale Month	Property Age	Price per SqFt
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
1	1340	5650	2014	10	71	188.0508475
2	1690	7639	2014	12	75	209.3385214
3	2720	8062	2015	2	93	233.7662338
4	1360	5000	2014	12	61	308.1632653
5	1800	7503	2015	2	39	303.5714286
6	4760	101930	2014	5	25	226.0147601
7	2238	6819	2014	6	31	150.1457726
8	1650	9711	2015	1	63	275.3301887
9	1780	8113	2015	4	66	128.9325843
10	2390	7570	2015	3	23	170.8994709
11	2210	8925	2015	4	61	186.0955056
12	1330	6000	2014	5	84	403.4482759
13	1780	12697	2014	5	99	216.7832168
14	1370	10208	2014	10	49	291.9708029
15	1360	4850	2015	3	126	292.8176796
16	2140	4000	2015	1	47	220.3389831
17	1890	14018	2014	7	32	208.994709
18	1610	4300	2014	5	110	303.125
19	1060	5095	2014	12	105	157.5
20	1280	8850	2015	4	57	184
21	1400	4980	2014	5	79	237.654321
22	4110	20336	2014	8	58	655.7377049
23						

Screenshot 5: Creating Total Area and Renovation Status

Custom columns were created for Total Area and Is Renovated. These columns provide additional information about property area and whether a property has been renovated. The explanation should match the formulas used in the dataset.

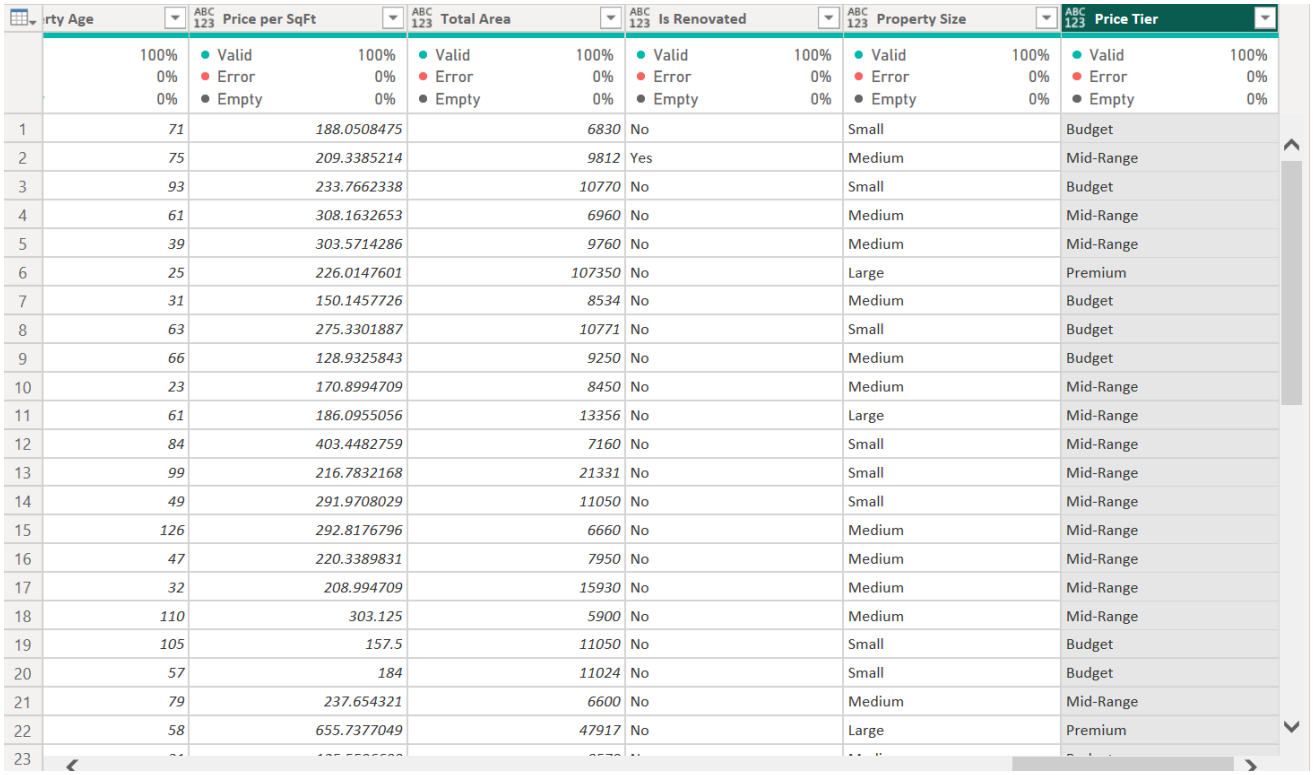


ear	Sale Month	Property Age	Price per SqFt	Total Area	Is Renovated
100% Valid	100% Valid	100% Valid	100% Valid	100% Valid	100% Valid
0% Error	0% Error	0% Error	0% Error	0% Error	0% Error
0% Empty	0% Empty	0% Empty	0% Empty	0% Empty	0% Empty
	10	71	188.0508475	6830	No
	12	75	209.3385214	9812	Yes
	2	93	233.7662338	10770	No
	12	61	308.1632653	6960	No
	2	39	303.5714286	9760	No
2014	5	25	226.0147601	107350	No
2014	6	31	150.1457726	8534	No
2015	1	63	275.3301887	10771	No
2015	4	66	128.9325843	9250	No
2015	3	23	170.8994709	8450	No
2015	4	61	186.0955056	13356	No
2014	5	84	403.4482759	7160	No
2014	5	99	216.7832168	21331	No
2014	10	49	291.9708029	11050	No
2015	3	126	292.8176796	6660	No
2015	1	47	220.3389831	7950	No
2014	7	32	208.994709	15930	No
2014	5	110	303.125	5900	No
2014	12	105	157.5	11050	No
2015	4	57	184	11024	No
2014	5	79	237.654321	6600	No
2014	8	58	655.7377049	47917	No

0 (0%) Error
0 (0%) Empty

Screenshot 6: Categorizing Property Size and Price Tier

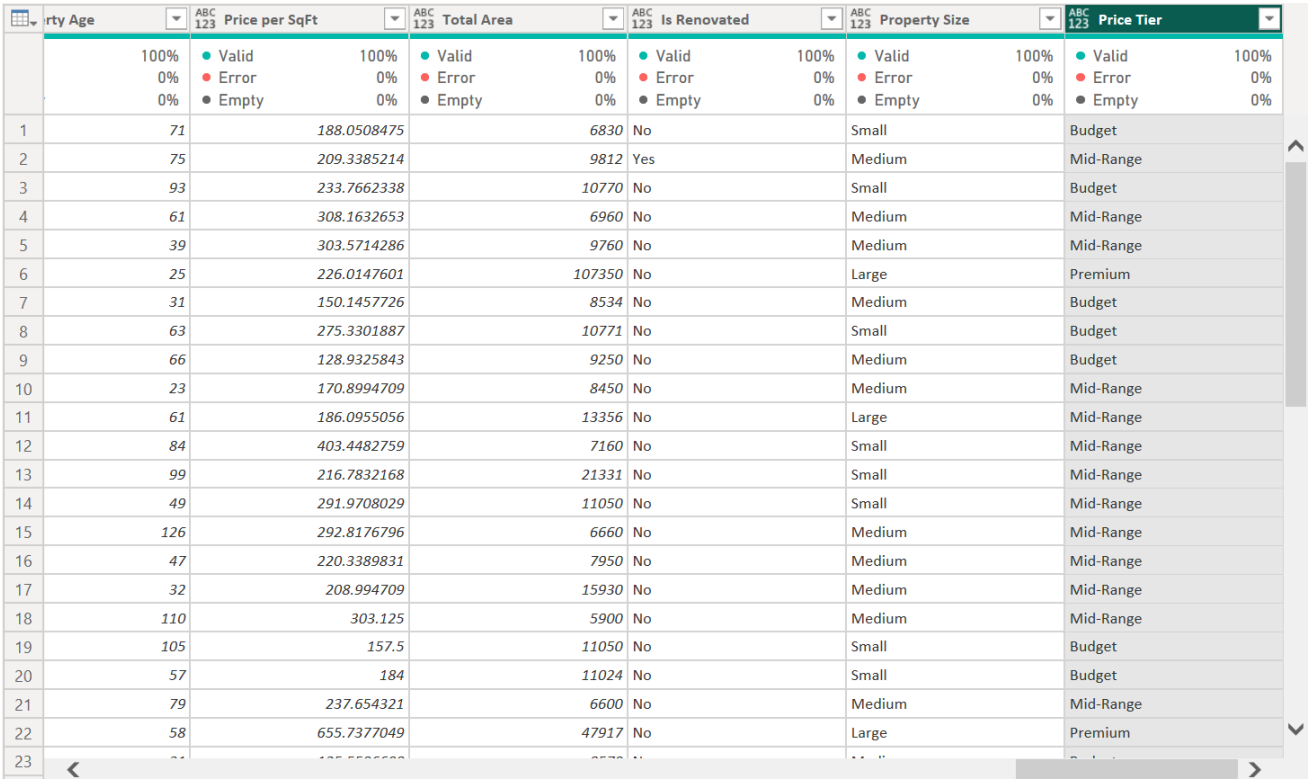
Properties were grouped into size categories such as Small, Medium, and Large, and price categories such as Budget, Mid-Range, and Premium. These categories make the data easier to compare.



	rty Age	Price per SqFt	Total Area	Is Renovated	Property Size	Price Tier
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
1	71	188.0508475	6830	No	Small	Budget
2	75	209.3385214	9812	Yes	Medium	Mid-Range
3	93	233.7662338	10770	No	Small	Budget
4	61	308.1632653	6960	No	Medium	Mid-Range
5	39	303.5714286	9760	No	Medium	Mid-Range
6	25	226.0147601	107350	No	Large	Premium
7	31	150.1457726	8534	No	Medium	Budget
8	63	275.3301887	10771	No	Small	Budget
9	66	128.9325843	9250	No	Medium	Budget
10	23	170.8994709	8450	No	Medium	Mid-Range
11	61	186.0955056	13356	No	Large	Mid-Range
12	84	403.4482759	7160	No	Small	Mid-Range
13	99	216.7832168	21331	No	Small	Mid-Range
14	49	291.9708029	11050	No	Small	Mid-Range
15	126	292.8176796	6660	No	Medium	Mid-Range
16	47	220.3389831	7950	No	Medium	Mid-Range
17	32	208.994709	15930	No	Medium	Mid-Range
18	110	303.125	5900	No	Medium	Mid-Range
19	105	157.5	11050	No	Small	Budget
20	57	184	11024	No	Small	Budget
21	79	237.654321	6600	No	Medium	Mid-Range
22	58	655.7377049	47917	No	Large	Premium
23						

Screenshot 7: Validating the Transformed Dataset

The final transformed dataset was checked in Power Query using column quality indicators. The visible indicators show the validity and quality of the displayed columns.



	rty Age	Price per SqFt	Total Area	Is Renovated	Property Size	Price Tier
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
1	71	188.0508475	6830	No	Small	Budget
2	75	209.3385214	9812	Yes	Medium	Mid-Range
3	93	233.7662338	10770	No	Small	Budget
4	61	308.1632653	6960	No	Medium	Mid-Range
5	39	303.5714286	9760	No	Medium	Mid-Range
6	25	226.0147601	107350	No	Large	Premium
7	31	150.1457726	8534	No	Medium	Budget
8	63	275.3301887	10771	No	Small	Budget
9	66	128.9325843	9250	No	Medium	Budget
10	23	170.8994709	8450	No	Medium	Mid-Range
11	61	186.0955056	13356	No	Large	Mid-Range
12	84	403.4482759	7160	No	Small	Mid-Range
13	99	216.7832168	21331	No	Small	Mid-Range
14	49	291.9708029	11050	No	Small	Mid-Range
15	126	292.8176796	6660	No	Medium	Mid-Range
16	47	220.3389831	7950	No	Medium	Mid-Range
17	32	208.994709	15930	No	Medium	Mid-Range
18	110	303.125	5900	No	Medium	Mid-Range
19	105	157.5	11050	No	Small	Budget
20	57	184	11024	No	Small	Budget
21	79	237.654321	6600	No	Medium	Mid-Range
22	58	655.7377049	47917	No	Large	Premium
23						

4. Conclusion

The Housing Data Cleaning and Transformation task was completed using Power BI Power Query. The dataset was prepared by extracting date attributes, creating custom columns, calculating property-related values, and categorizing properties by size and price tier. The transformed dataset provides a structured foundation for further analysis of house sales and property prices.